

Gatik Yadav

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EDUCATION

Jaypee University of Engineering and Technology
Bachelor of Technology in Computer Science Engineering; CGPA: 7.1/10

Aug 2023 – May 2027
Guna, MP

EXPERIENCE

Full Stack Developer Intern

CollegeTips.in

May 2025 – July 2025

Bhopal, Madhya Pradesh

- Developed and deployed a full-stack web application integrating an AI-powered multilingual chatbot.
- Designed conversational workflows covering **25+ digital literacy intents**, including WhatsApp usage and UPI payment apps.
- Achieved approximately **90% intent-matching accuracy** during controlled testing using structured NLP prompts.
- Optimized backend request handling to deliver chatbot responses in **under 1.2 seconds** on average.
- Implemented Hindi and English language support, improving accessibility for non-English users.

PROJECTS

Full Stack Media Gallery Platform | *HTML5, CSS, JavaScript, Node.js, Express.js, MongoDB*

May 2025

- Built and deployed a full-stack gallery application enabling user authentication, profile management, and secure media uploads.
- Designed a scalable backend using MongoDB, supporting **100+ images/videos per user** with structured metadata storage.
- Integrated Cloudinary for media hosting, achieving average upload times of **4–5 seconds** for files up to **10 MB**.
- Implemented **JWT-based authentication** and a secure email password-reset workflow using Nodemailer (Gmail SMTP).
- Enforced protected routes and server-side validation to ensure data integrity and application security.
- **Live Demo:** virtual-gallery-app.vercel.app | **GitHub:** github.com/Gatik8205/Virtual-Gallery-App

Face Recognition System | *Python, Dlib, OpenCV*

December 2025

- Developed a real-time face recognition system using computer vision and deep feature encoding techniques.
- Achieved approximately **92%** face recognition accuracy on a custom dataset of 200+ images across 20 identities.
- Optimized the detection and recognition pipeline to identify faces in under 1.5 seconds per frame.
- Implemented known/unknown face classification, reducing false positives by around **30%**.
- Supported real-time webcam processing at 15–20 FPS under standard lighting conditions.
- **GitHub:** github.com/Gatik8205/smart_surveillance_system

Document Fraud Detection Using AI | *React, Python, FastAPI, CNN*

Ongoing

- Built an AI-Based Document Fraud Detection System using a hybrid pipeline combining ResNet18 (CNN) with classical forensics ELA, Canny Edge Detection, ORB Clone Detection, Metadata Analysis, and OCR, achieving **72%** validation accuracy on 24,000+ images across CASIA 2.0, CG-1050, and COMOFOD datasets.
- Implemented Grad-CAM visualization on ResNet18's final convolutional layer to produce interpretable heatmaps, making CNN predictions explainable by highlighting tampered regions in uploaded documents.
- Designed a weighted hybrid scoring engine **45% image · 30% metadata · 25% text** with confidence calibration using score variance, outputting Low Risk / Suspicious / High Risk decisions with High / Medium / Low confidence levels.
- Built a full-stack deployment with a React (TypeScript) + Tailwind CSS frontend and FastAPI backend, featuring multi-page PDF support via PyMuPDF, Grad-CAM heatmap rendering, and automated PDF report generation — deployed live on Vercel + Render.
- **GitHub:** github.com/Gatik8205/Document_Fraud_Detection

TECHNICAL SKILLS

Languages: JavaScript, Python, HTML5, CSS3

Frontend: React.js, Tailwind CSS, Responsive Design

Backend: Node.js, Express.js, JWT Authentication, FASTAPI

Database: MongoDB

AI/ML: Dlib, OpenCV, CNN

Tools: Git, GitHub, Postman, VS Code, Render, Vercel

CERTIFICATIONS & ACHIEVEMENTS

ISRO Certificate of Merit: Completed Space Science and Technology Awareness Training (START) program

Data Analytics Job Simulation (Deloitte): Completed a Deloitte job simulation involving data analysis and forensic technology

Supervised Machine Learning: Regression and Classification (Coursera)